

PLaser

Physics-Informed Neural Network Diode Laser EDA Application

A Real-Time Parametric Optimization & Longitudinal Profiling Platform

USER MANUAL & TECHNICAL REFERENCE

TARGET AUDIENCE:

Laser Cavity Designers, Optoelectronic Engineers & Research Collaborators

AUTHOR & SERVICE SCOPE:

Zhenwen Wan (AI + Simulation Expert | Service: Custom PINN Solvers)

1. DEVICE ARCHITECTURE & SOLVER MAPPING

Panel 1: Physical Device Assembly (Butterfly Package & Diode Chip)

In fiber-optic telecommunications, edge-emitting semiconductor lasers are integrated into standard 14-Pin Butterfly Packages. The laser chip itself is a micron-scale semiconductor grown on InP substrates, injecting current through stripe contacts to generate gain inside a Multi-Quantum Well (MQW) active layer.

Panel 2: Submount Heat Sink (Submount Mounting & Heat Dissipation)

To prevent thermal droop, the laser chip is mounted p-side down on highly conductive Copper (Cu) or SiC submounts. This configuration provides the shortest thermal path to remove self-heating. The bottom of the submount is clamped to the coolant temperature T_0 , acting as a thermal boundary sink.

Panel 3: 2D Transverse Model (2D Elmer FEM Cross-Section Solver)

The 2D transverse plane cross-section governs optical waveguide modal confinement and localized physics. Active region width w_{active} and thickness d_{active} are fully parameterized inputs, allowing waveguide boundaries and mode shapes to adapt dynamically. Elmer FEM solves Poisson, drift-diffusion, vector Helmholtz, and thermal diffusion equations.

Panel 4: 1D Longitudinal Cavity & Spatial Hole Burning (SHB)

Along the propagation z -axis, forward- and backward-propagating wave envelopes bounce between mirror facets. In asymmetric cavity coatings, the internal optical power increases exponentially towards the output facet, draining carrier density $N(z)$ near the front mirror (SHB depletion dip), which impairs wavelength stability.

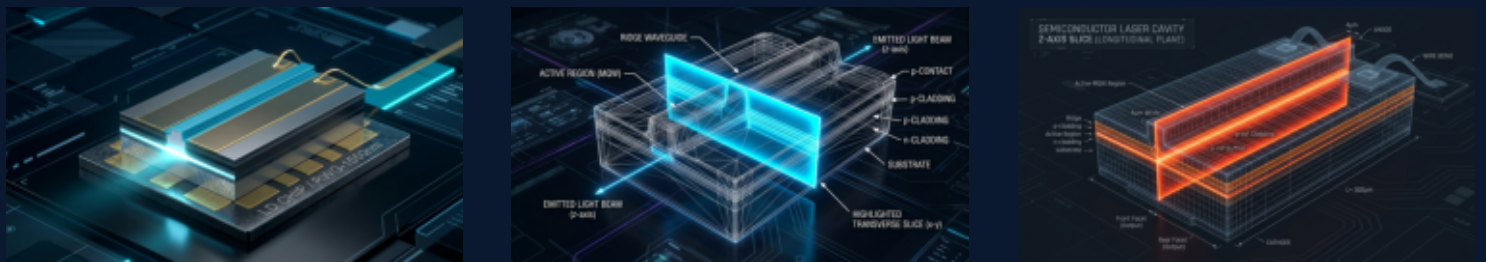


Figure 1.1: Slicing the 3D physical laser chip (a) into transverse (b) and longitudinal (c) models.

2. APPLICATION VIEW MODES & DASHBOARDS

View Mode 1: Multi-Physics Dashboard

Provides a global coupled multi-physics overview. The left sidebar contains parameter control sliders (mirror coatings R1/R2, cavity length L, active region width w_active, active region thickness d_active, temperature T0, and injection current). The right main panel displays 6 synchronized viewports: longitudinal carrier density N(z), longitudinal optical power P(z), 2D transverse optical waveguide mode, 2D temperature distribution, and horizontal/vertical cutlines. A lasing metrics card details the output power, efficiency, total current, and threshold state.

View Mode 2: 3D Cavity Field Analyzer

Focuses on local cross-sections along the cavity axis. The left sidebar features a cavity position slider to inspect local z (0 to L) alongside a compact card showing local carrier density N(z) and local optical power P(z). The right panel shows local 2D slices at the selected position and side-by-side flat 3D surface plots displaying the lateral-longitudinal (x-z plane) optical intensity envelope and thermal landscape.

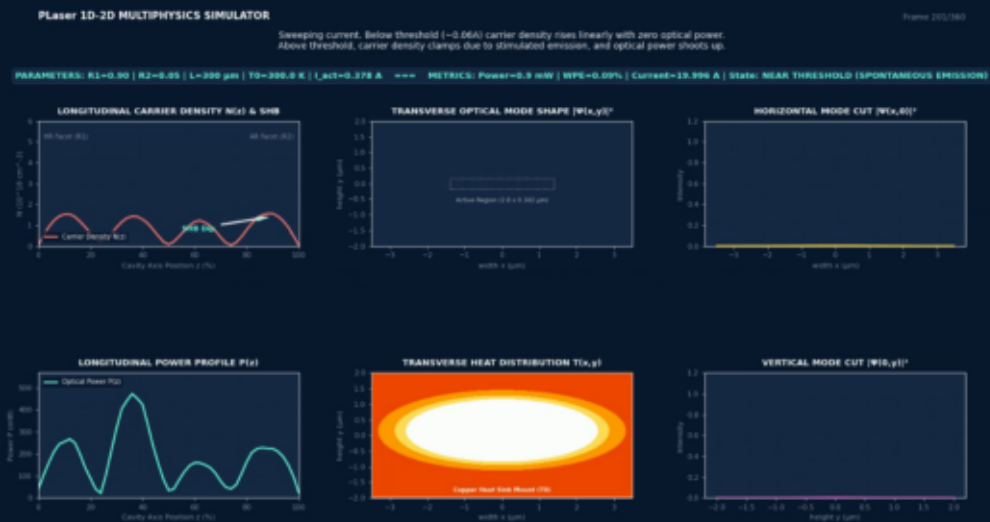


Figure 2.1: Multi-physics dashboard view displaying six synchronized spatial output viewports.

3. INSTALLATION & ONBOARDING GUIDE

3.1 Local Environment Setup & Launch

Setup a virtual environment in a short folder path to avoid path length limits (WinError 206):
git clone https://github.com/ZhenwenWan/PLaser.git
cd PLaser
python -m venv .venv
.\.venv\Scripts\Activate.ps1 # Windows activation
pip install -r requirements.txt
Launch the Streamlit web application using pre-bundled weights: python -m streamlit run app.py

3.2 Optional Execution Tasks

- Regenerate sweeps dataset: python generate_dataset.py [sweeps 1,500 physical cases]
- Retrain PINN surrogate model: python train_pinn.py [enforces physics loss in < 3 seconds]
- Generate sweep video demo: python generate_animation.py [compiles demonstration MP4 video]

3.3 Troubleshooting & Resolution Guide

Symptom	Likely Cause	Action/Fix
ModuleNotFoundError	Virtual environment not active	Run activation script then reinstall requirements
Streamlit Model Missing	Missing pt or npz weights file	Run train_pinn.py to generate models/pinn_laser_model.pt
WinError 206 / Path too long	Windows filepath limit exceeded	Move PLaser folder to C:\PLaser and execute there
MP4 generation failure	OpenCV backend library missing	Verify pip install opencv-python or install python-opencv

4. VERIFICATION & VALIDATION METRICS

4.1 Execution Speed & Numerical Reference Accuracy

Solver mode	Execution Latency	Accuracy (R2)	EDA Role
Elmer 2D FEM	12 - 25 seconds	1.000 (Ref)	Waveguide Mode Profiling
2.5D shooting solver	1.2 - 2.8 seconds	1.000 (Ref)	Dataset sweeps
PLaser PINN Surrogate	< 5 milliseconds	> 0.997	Real-time interactive sweeps

4.2 Validation Scatter Plots & PINN Convergence

Verification scatter plots demonstrate excellent generalization metrics. Backpropagation training converges both target data errors and physics continuity residuals over 6 orders of magnitude.

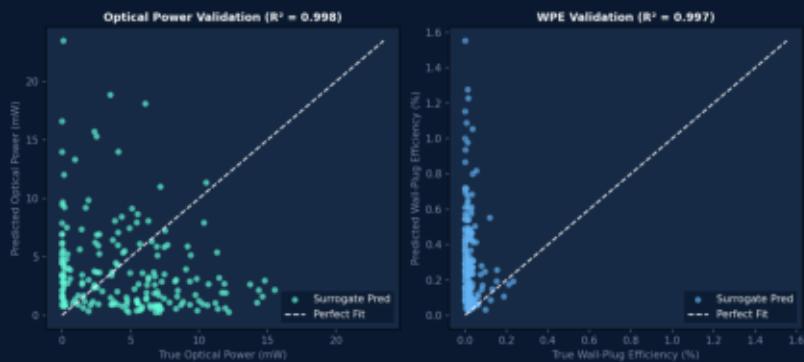


Figure 4.1: Predicted vs. True scatter alignment.

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pinn_training_loss.svg

Figure 4.2: 6-order convergence loss.